_eda_price_2025_Q4

March 31, 2026

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import subprocess
import os
from scipy.stats import chi2_contingency
from datetime import timedelta

from etl.customer import aggregate_customer
from etl.subsidy import report_customer_conversion_by_subsidy_issue_status,␣
↪report_customer_metrics_by_subsidy_issue_status
from etl.subsidy import visualize_conversion_by_subsidy_issues,␣
↪report_customer_subsidy_issues_by_product
from etl.subsidy import visualize_customer_subsidy_issues_by_product,␣
↪visualize_policy_suspension_impact
from etl.price import visualize_conversion_by_price_sweet_spots,␣
↪visualize_conversion_by_price_sweet_spots_variant
from etl.price import report_product_conversion_by_price_quartile,␣
↪visualize_conversion_by_equipment_category_price
from etl.price import visualise_conversion_by_brand_price,␣
↪show_two_panel_conversion_by_brand_price
from etl.price import visualize_conversion_by_equipment_type_brand_price,␣
↪show_conversion_by_price_dashboard,␣
↪show_product_type_price_conversion_heatmap
from etl.season import visualize_conversion_by_season,␣
↪visualize_regional_seasonality
from etl.period import visualize_conversion_by_year,␣
↪report_best_and_worst_months, show_conversion_by_price_over_time,␣
↪overlay_all_years_on_one_plot
from etl.period import visualize_conversion_by_prices_sweet_spot_shift,␣
↪visualize_decision_times, show_decision_time_dashboard
from etl.product import visualise_heat_pump_performance,␣
↪show_product_type_by_brand_heatmap, show_product_type_by_agency_heatmap,␣
↪show_brand_by_product_type_heatmap
```

```python
import warnings
warnings.filterwarnings('ignore')

df_clean = pd.read_csv('cleaned_quote_data.csv')
df_clean['dt_creation_devis'] = pd.to_datetime(df_clean['dt_creation_devis'])
print(f"Loaded data: {len(df_clean):,} quotes from {df_clean['numero_compte'].
 ↪nunique():,} customers")
```

```
Loaded data: 37,008 quotes from 25,716 customers
```

```python
#customers = aggregate_customer(df_clean)
```

```python
all_customers = pd.read_csv('customer_master_data.csv')
```

```python
# Using quarter column (if you have it)
customers_2025_q4 = all_customers[
    (all_customers['year'] == 2025) &
    (all_customers['quarter'] == 4)
]

# Using date range
customers_2025_q4 = all_customers[
    (all_customers['first_quote_date'] >= '2025-10-01') &
    (all_customers['first_quote_date'] <= '2025-12-31')
]

print(f"Q4 2025 customers: {len(customers_2025_q4):,}")
```

```
Q4 2025 customers: 2,872
```

```python
customers = customers_2025_q4
```

```python
conv_no_issue, conv_with_issue =␣
 ↪report_customer_conversion_by_subsidy_issue_status(customers)
report_customer_metrics_by_subsidy_issue_status(customers)
visualize_conversion_by_subsidy_issues(customers, conv_no_issue,␣
 ↪conv_with_issue)
visualize_policy_suspension_impact(customers, price_var='max_out_of_pocket')

top_products, no_issue_pct, issue_pct =␣
 ↪report_customer_subsidy_issues_by_product(customers)
visualize_customer_subsidy_issues_by_product(customers, top_products,␣
 ↪no_issue_pct, issue_pct)

visualize_conversion_by_price_sweet_spots(customers,␣
 ↪price_var='max_out_of_pocket')
```

```
visualize_conversion_by_price_sweet_spots_variant(customers,␣
 ↪price_var='max_out_of_pocket')
report_product_conversion_by_price_quartile(customers)
visualize_conversion_by_equipment_category_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualise_conversion_by_brand_price(customers, price_var='max_out_of_pocket')
show_two_panel_conversion_by_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_conversion_by_equipment_type_brand_price(customers,␣
 ↪price_var='max_out_of_pocket')

visualize_conversion_by_season(customers, price_var='max_out_of_pocket')
visualize_regional_seasonality(customers, price_var='max_out_of_pocket')

visualize_conversion_by_year(customers, price_var='max_out_of_pocket')
report_best_and_worst_months(customers, price_var='max_out_of_pocket')
show_conversion_by_price_over_time(customers, price_var='max_out_of_pocket')
overlay_all_years_on_one_plot(customers, price_var='max_out_of_pocket')
visualize_conversion_by_prices_sweet_spot_shift(customers,␣
 ↪price_var='max_out_of_pocket')
visualize_decision_times(customers, price_var='max_out_of_pocket')
show_product_type_price_conversion_heatmap(customers,␣
 ↪price_var='max_out_of_pocket')
show_decision_time_dashboard(customers, price_var='max_out_of_pocket')

visualise_heat_pump_performance(customers, price_var='max_out_of_pocket')

show_conversion_by_price_dashboard(customers, price_var='max_out_of_pocket')

show_product_type_by_brand_heatmap(customers)
show_product_type_by_agency_heatmap(customers)
show_brand_by_product_type_heatmap(customers)
```

```
Customer conversion by subsidy issue status:

                   conversion_rate  customer_count  converted_count
had_subsidy_issue
False                     0.333853            1923            642.0
True                      0.419389             949            398.0

Chi-square p-value: 8.790733e-06
Statistically significant: YES

Conversion without issues: 33.39%
Conversion with issues: 41.94%
Difference: 8.6 percentage points
```

```
Customer-level finding: +8.6 points

================================================================================
Customer characteristics by subsidy issue status
================================================================================

Metric comparison:

total_quotes:
  No issues: 1.35
  With issues: 1.53
  Ratio: 1.13x

avg_quote_amount:
  No issues: 5644.64
  With issues: 10317.63
  Ratio: 1.83x

max_quote_amount:
  No issues: 5992.35
  With issues: 11087.40
  Ratio: 1.85x

customer_duration_days:
  No issues: 4.21
  With issues: 7.05
  Ratio: 1.67x

price_range:
  No issues: 767.85
  With issues: 1454.56
  Ratio: 1.89x

multiple_quotes:
  No issues: 0.27
  With issues: 0.36
  Ratio: 1.35x

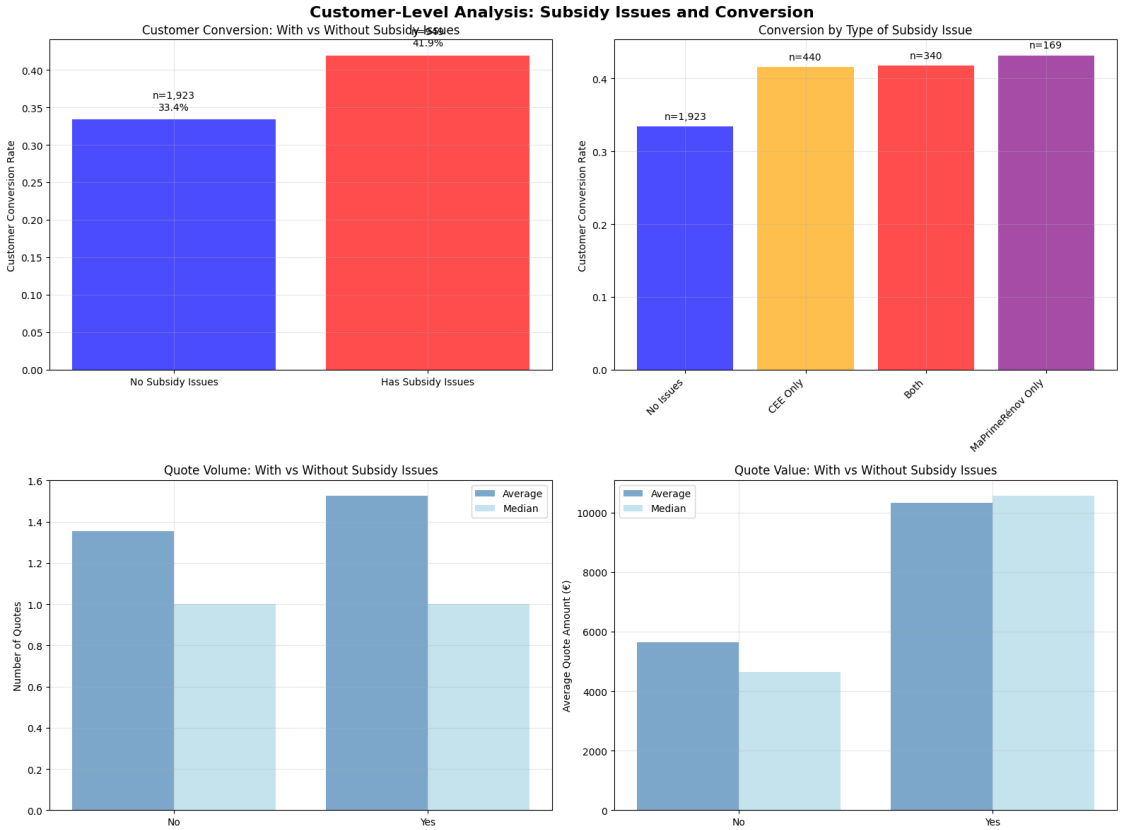
Quote pattern distribution by subsidy issue:
had_subsidy_issue  False  True
quote_pattern
Single quote       0.733  0.641
Three+ quotes      0.057  0.110
Two quotes         0.210  0.250

Decision speed by subsidy issue:
had_subsidy_issue   False  True
decision_speed
```

```
Long (>30 days)     0.038  0.074
Medium (8-30 days)  0.034  0.052
Quick (2-7 days)    0.040  0.064
Same day decision   0.888  0.810

Product family by subsidy issue status:
had_subsidy_issue                 False  True
main_product_family
Accessoire de pose                0.002  0.001
Appareil hybride                  0.004  0.046
Autres                            0.012  0.006
Chaudière                         0.457  0.051
Climatisation                     0.212  0.092
ECS : Chauffe-eau ou adoucisseur  0.071  0.019
Emetteur de chauffage  ou chappe  0.009  0.007
Photovoltaïque                    0.012  0.012
Plomberie Sanitaire               0.001  0.000
Pompe à chaleur                   0.041  0.468
Poêle                             0.168  0.298
Produit VMC                       0.006  0.000
Unknown                           0.005  0.000

Conversion by subsidy issue type:
                        mean  count
subsidy_issue_type
MaPrimeRénov Only   0.431953    169
Both                0.417647    340
CEE Only            0.415909    440
No Issues           0.333853   1923
```

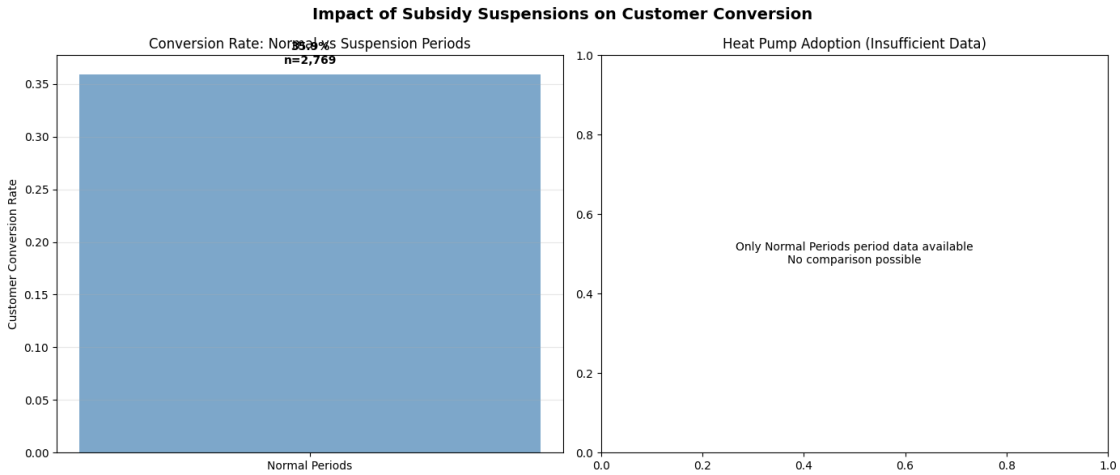



```
================================================================================
Policy Suspension Impact
================================================================================

Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263
Unique values in during_suspension: [False]

   Only data during normal periods available

Conversion during subsidy suspensions:
                    mean  count
Normal Periods  0.359336   2769
```

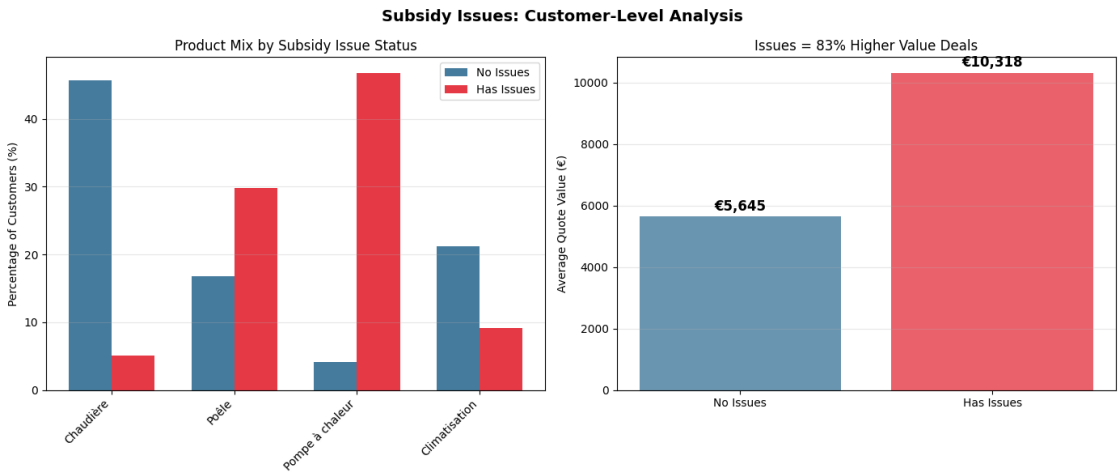



```
Unique values in main_product_family:
main_product_family
Chaudière                          926
Poêle                              607
Pompe à chaleur                    523
Climatisation                      495
ECS : Chauffe-eau ou adoucisseur   154
Appareil hybride                    52
Photovoltaïque                      34
Autres                              30
Emetteur de chauffage  ou chappe    25
Produit VMC                         12
Unknown                              9
Accessoire de pose                   4
Plomberie Sanitaire                  1
Name: count, dtype: int64

Top products:
main_product_family
Chaudière                          926
Poêle                              607
Pompe à chaleur                    523
Climatisation                      495
ECS : Chauffe-eau ou adoucisseur   154
Appareil hybride                    52
Photovoltaïque                      34
Autres                              30
Emetteur de chauffage  ou chappe    25
Produit VMC                         12
Name: count, dtype: int64
```

```
Using these products: ['Chaudière', 'Poêle', 'Pompe à chaleur', 'Climatisation']

No Issue %: [np.float64(45.657826313052524), np.float64(16.848673946957877),
np.float64(4.108164326573062), np.float64(21.21684867394696)]
Has Issue %: [np.float64(5.05795574288725), np.float64(29.820864067439413),
np.float64(46.78609062170706), np.float64(9.167544783983141)]
```

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================================================================================
Threshold effect
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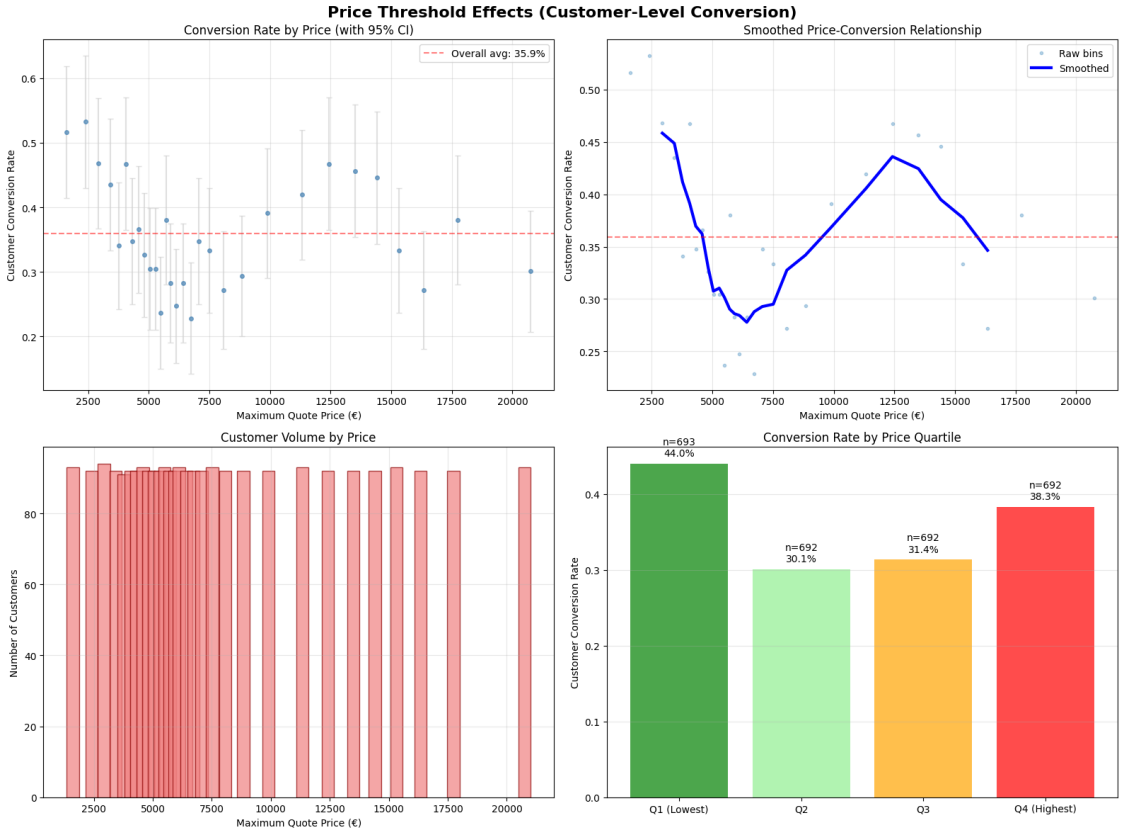
Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263
```

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================================================================================
THRESHOLD EFFECT: KEY FINDINGS
================================================================================

Conversion by Price Quartile:
  Q1 (Lowest): 44.0% (n=693)
  Q2: 30.1% (n=692)
  Q3: 31.4% (n=692)
  Q4 (Highest): 38.3% (n=692)

Difference between lowest and highest quartile: 5.7 percentage points

Sweet spots (above-average conversion):
  €1,611: 51.6% (n=93)
  €2,410: 53.3% (n=92)
  €2,935: 46.8% (n=94)
  €3,431: 43.5% (n=92)
  €4,065: 46.7% (n=92)
  €4,575: 36.6% (n=93)
  €5,715: 38.0% (n=92)
```

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  €9,905: 39.1% (n=92)
  €11,336: 41.9% (n=93)
  €12,425: 46.7% (n=92)
  €13,500: 45.6% (n=92)
  €14,421: 44.6% (n=92)
  €17,733: 38.0% (n=92)

Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263

================================================================================
Product mix by price quartile
================================================================================

Product distribution by price quartile (%):
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
price_quartile
Q1_Low                   21.2    44.6        0.1   24.6    9.5
Q2_MidLow                14.6    39.6        0.4    2.4   43.0
Q3_MidHigh               22.9    37.3        7.2    3.4   29.2
Q4_High                  13.0     9.5       67.9    8.6    1.0

================================================================================
Conversion rates by product and price quartile
================================================================================

Heat Pump (n=554):
  Q1_Low: 0.0%
  Q2_MidLow: 33.3%
  Q3_MidHigh: 58.8%
  Q4_High: 42.7%

Boiler (n=925):
  Q1_Low: 47.9%
  Q2_MidLow: 31.1%
  Q3_MidHigh: 33.1%
  Q4_High: 25.4%

AC (n=509):
  Q1_Low: 40.0%
  Q2_MidLow: 31.1%
  Q3_MidHigh: 29.6%
  Q4_High: 27.2%

Stove (n=606):
  Q1_Low: 37.3%
  Q2_MidLow: 29.6%
```

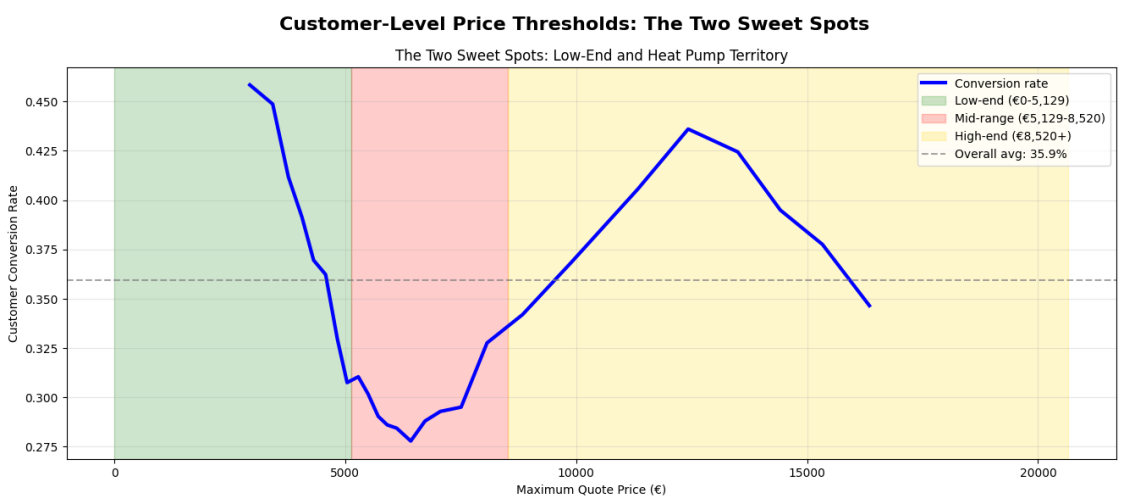
```
  Q3_MidHigh: 24.3%
  Q4_High: 71.4%

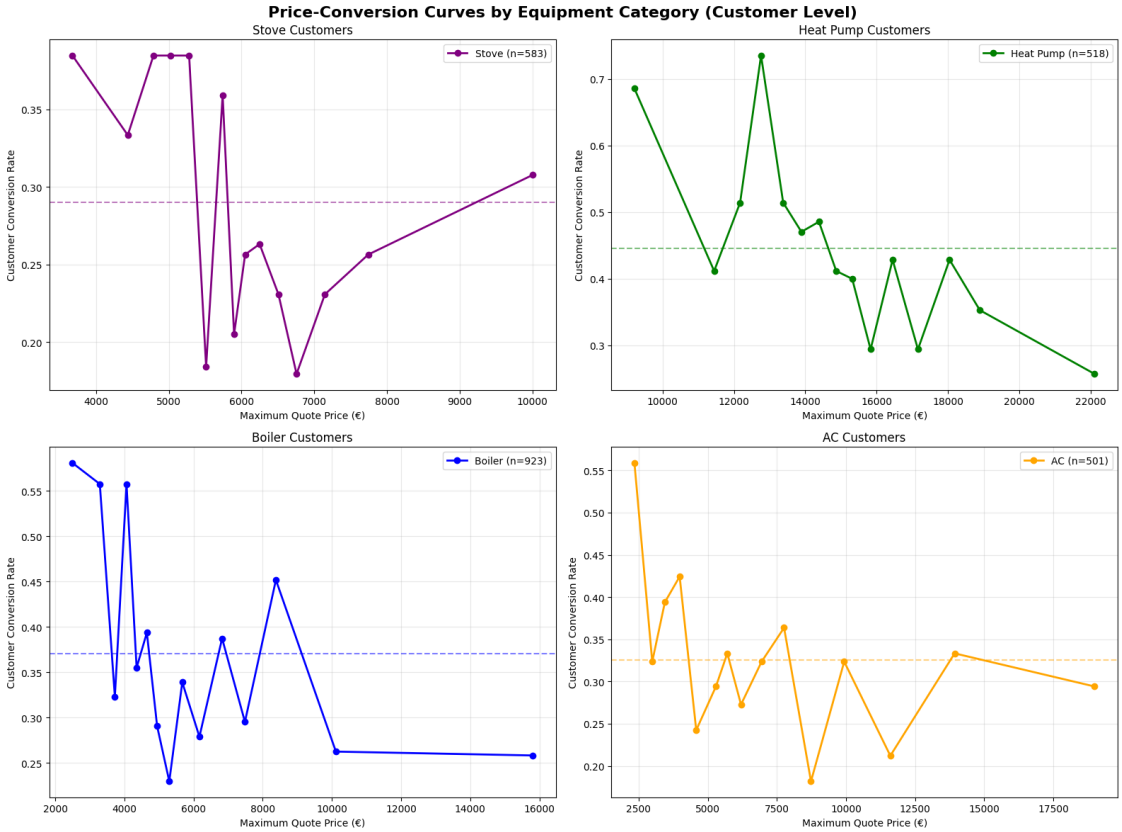
================================================================================
HEAT PUMP CUSTOMERS (n=633)
================================================================================
Average price: €15,786
Conversion rate: 43.3%
Subsidy issue rate: 84.8%

Heat pump adoption by quartile:
  Q1_Low: 0.3%
  Q2_MidLow: 0.4%
  Q3_MidHigh: 8.9%
  Q4_High: 77.2%

================================================================================
Equipment Category Segment Effect
================================================================================

Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263
```



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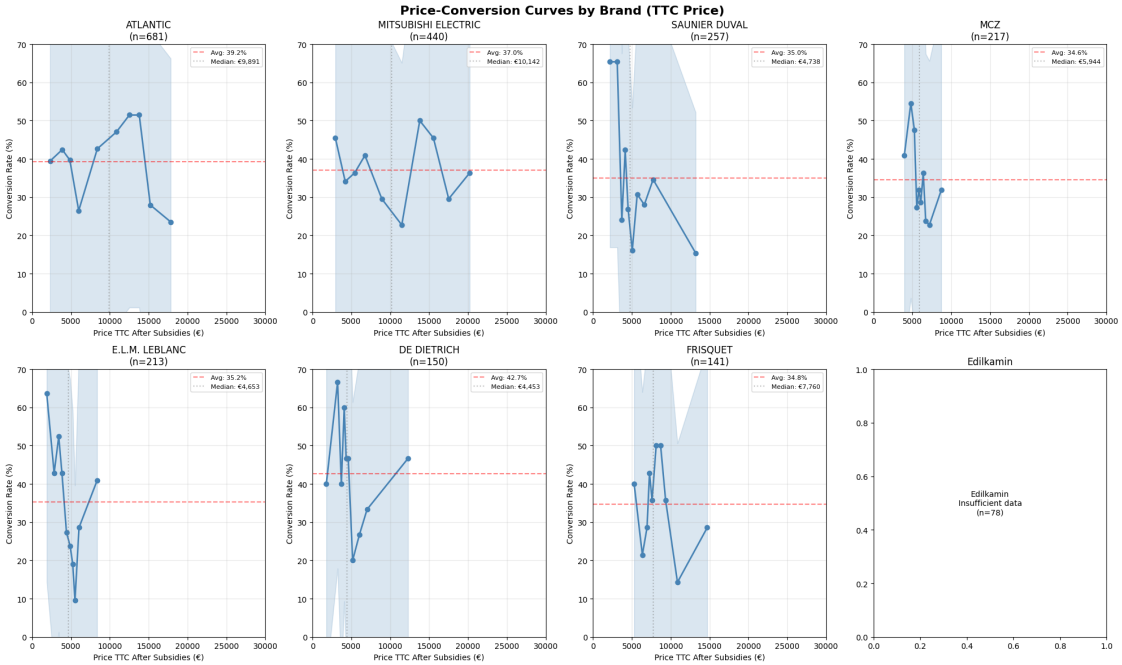
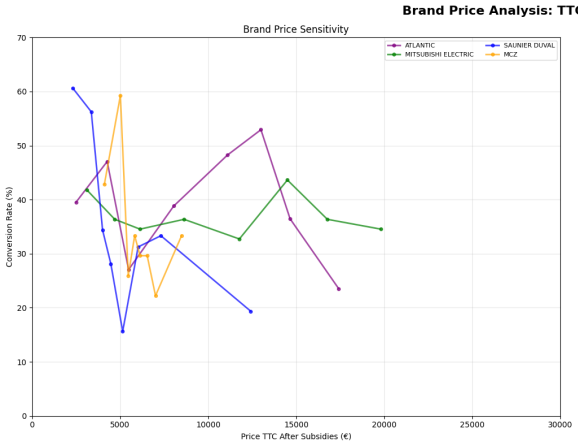
================================================================================
Conversion by Brand Price
================================================================================

Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263
Top 8 brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC', 'SAUNIER DUVAL', 'MCZ',
'E.L.M. LEBLANC', 'DE DIETRICH', 'FRISQUET', 'Edilkamin']
```




```
Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263
```



Brand Performance Summary

Brand	Customers	Avg Conv	Avg Price	Best Price Range	Best Conv
ATLANTIC	681	39.2%	€9,544	€11,835-14,475	51.5%
MITSUBISHI ELECTRIC	440	37.0%	€10,677	€12,705-16,482	47.7%
SAUNIER DUVAL	257	35.0%	€5,599	€1,343-3,420	65.4%
MCZ	217	34.6%	€6,063	€1,800-5,071	47.7%
E.L.M. LEBLANC	213	35.2%	€4,642	€1,347-3,279	53.5%
DE DIETRICH	150	42.7%	€5,214	€1,371-3,559	53.3%
FRISQUET	141	34.8%	€8,497	€7,441-8,460	42.9%

```
================================================================================
TTC Price-conversion curves: 4 equipment types, multiple brands per graph
================================================================================

Analyzing 2,769 customers
```

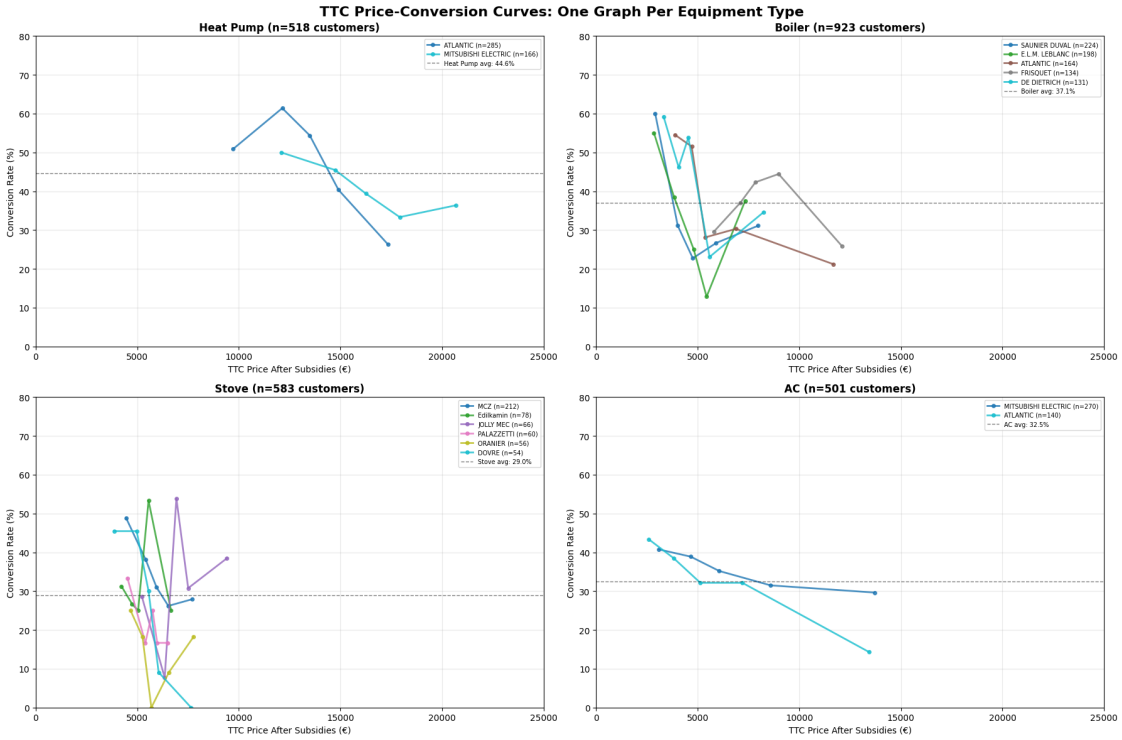
```
Price range: €1,343 to €25,821
Median price: €6,263

Heat Pump - Top brands: ['ATLANTIC', 'MITSUBISHI ELECTRIC']

Boiler - Top brands: ['SAUNIER DUVAL', 'E.L.M. LEBLANC', 'ATLANTIC', 'FRISQUET',
'DE DIETRICH']

Stove - Top brands: ['MCZ', 'Edilkamin', 'JOLLY MEC', 'PALAZZETTI', 'ORANIER',
'DOVRE']

AC - Top brands: ['MITSUBISHI ELECTRIC', 'ATLANTIC']
```

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================================================================================
Seasonal Effects on Customer Conversion
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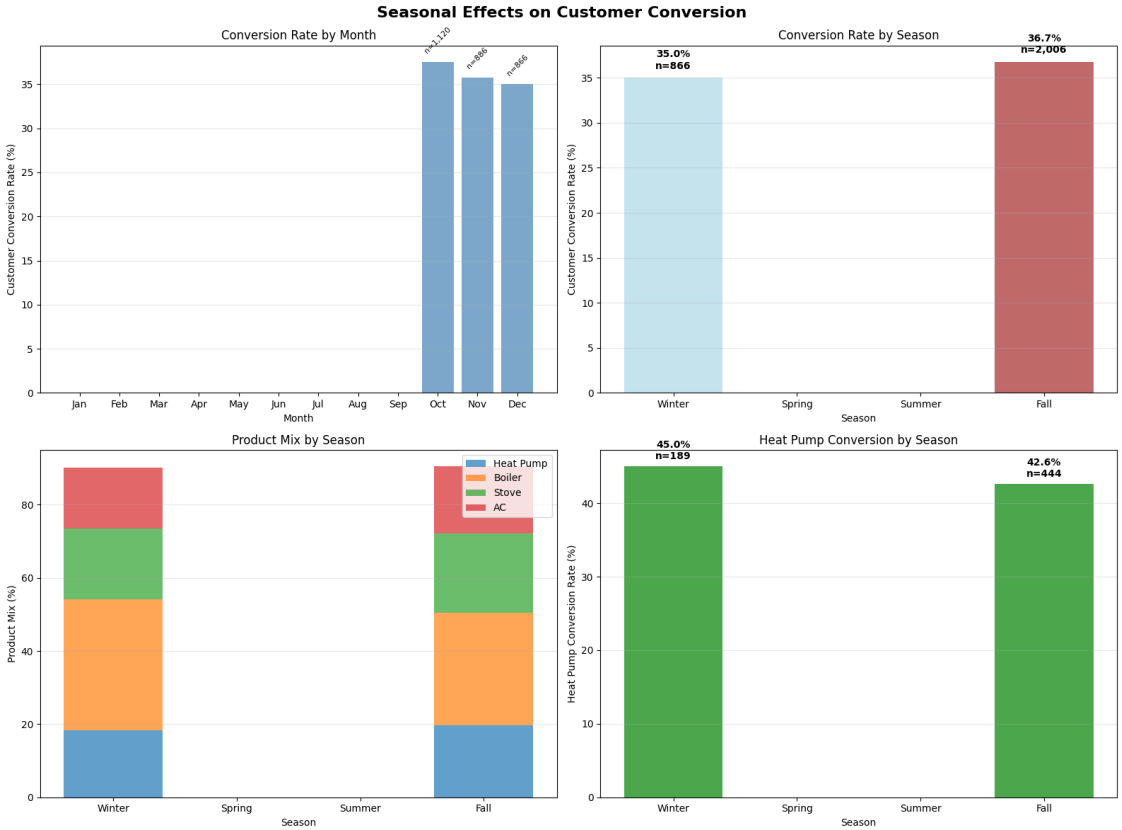


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================================================================================
SEASONAL STATISTICS
================================================================================

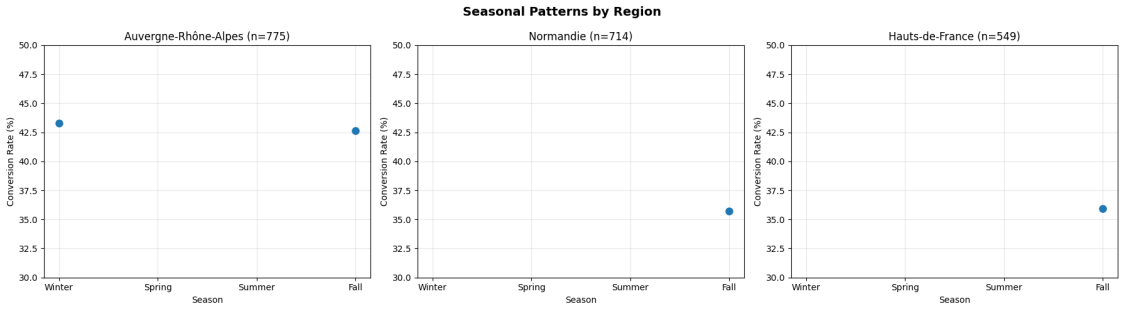
Conversion by season:
  Winter: 35.0% (n=866)
  Fall: 36.7% (n=2,006)

Seasonality chi-square p-value: 0.3932
Statistically significant: NO

================================================================================
Regional Seasonal Pattern
================================================================================

Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263

Top regions: ['Auvergne-Rhône-Alpes', 'Normandie', 'Hauts-de-France']
```

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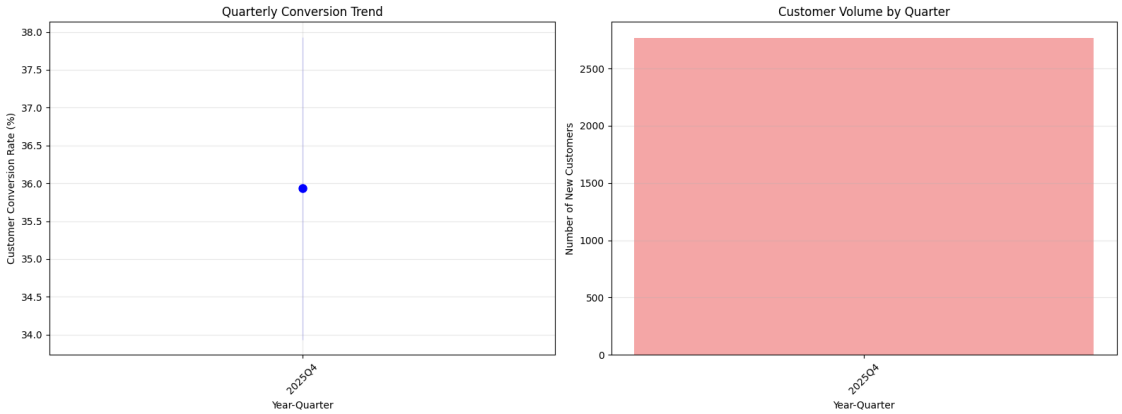
================================================================================
Year-over-year trends
================================================================================

Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263

Quarterly conversion rates:
                  mean  count
year_quarter
2025Q4        0.359336   2769
```

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================================================================================
Best and Worst Months Report
================================================================================

Analyzing 2,769 customers
Price range: €1,343 to €25,821
```

```
Median price: €6,263

  Only one suspension period found: [False]
Skipping suspension comparison

Best month: 10 (37.6%, n=1080)

Best season: Fall (36.5%)
Worst season: Winter (34.6%)

Heat pump conversion by season:
  Winter: 44.9%
  Fall: 43.8%

Analyzing 2,769 customers
Price range: €1,343 to €25,821
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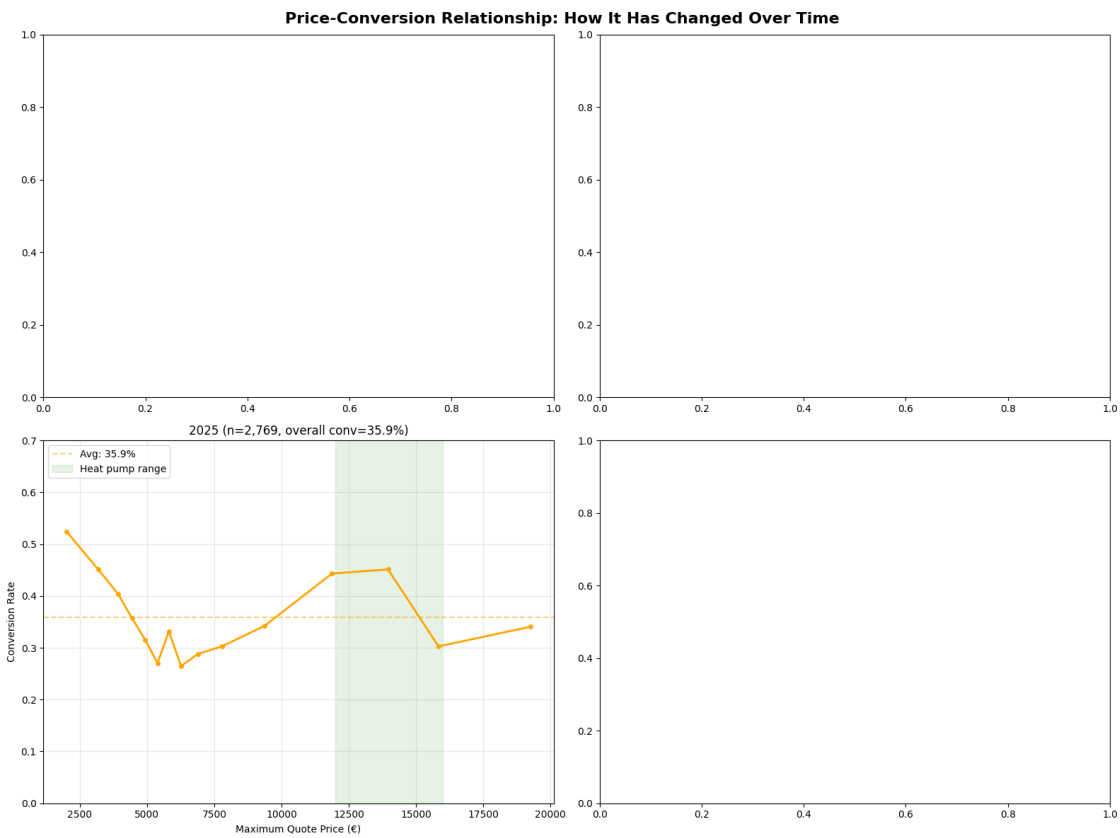
```

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Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263
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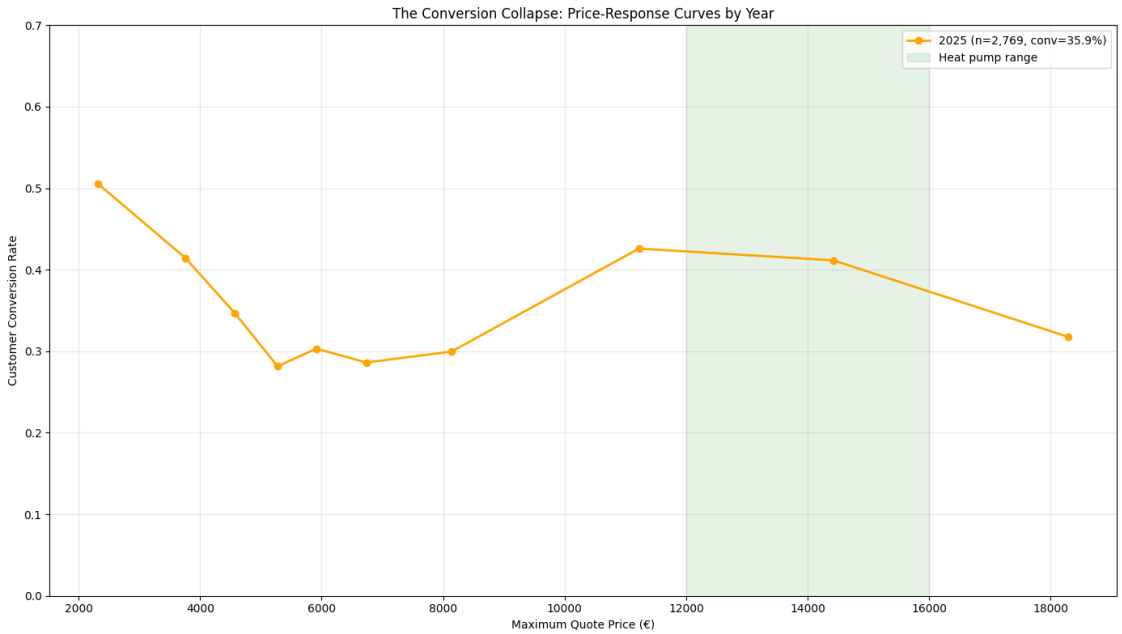


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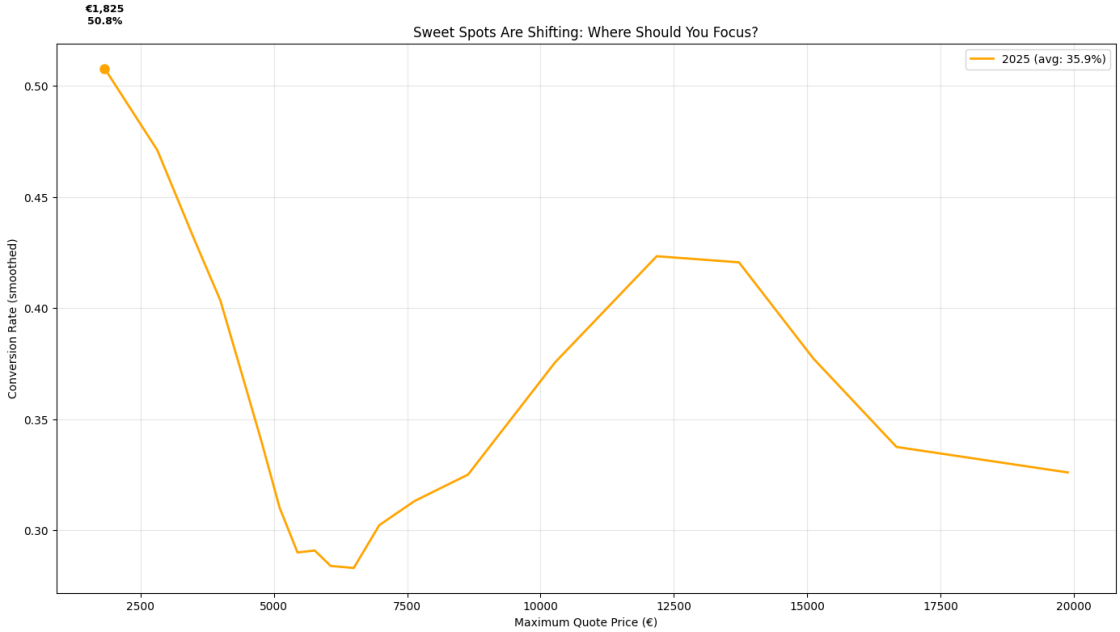
================================================================================
Sweet Spot Analysis Over Time
================================================================================

Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263

Sweet spots by year (price points with conversion > avg + 3%):

2025 (overall avg: 35.9%):
  €1,825: 53.2% (n=139)
  €2,812: 47.9% (n=140)
  €3,516: 39.4% (n=137)
  €3,993: 43.5% (n=138)
  €10,273: 39.6% (n=139)
  €12,179: 45.7% (n=138)
  €13,721: 44.9% (n=138)
```

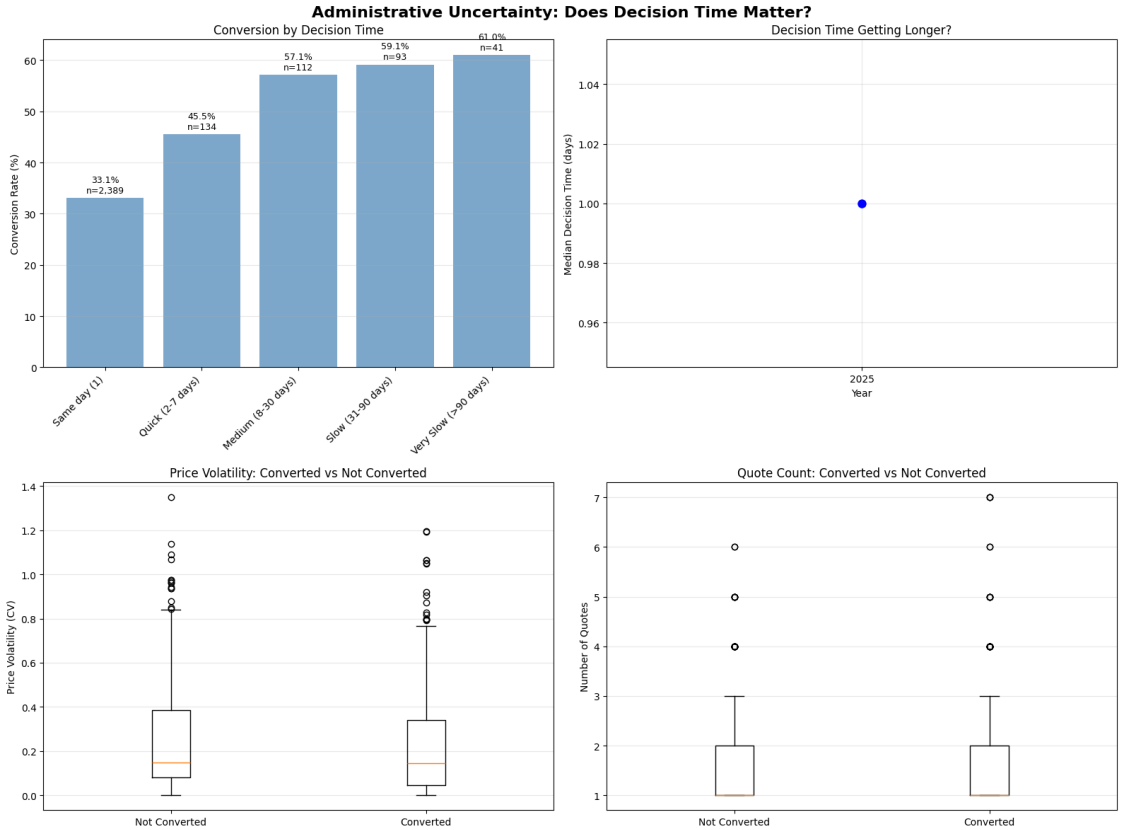



```
================================================================================
Administrative Uncertainty Effect
================================================================================

Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263

Decision time distribution:
count    2769.000000
mean        5.037559
std        16.320845
min         1.000000
25%         1.000000
50%         1.000000
75%         1.000000
max       145.000000
Name: decision_days, dtype: float64

Customers with 1-day decision: 2,389 (86.3%)
```

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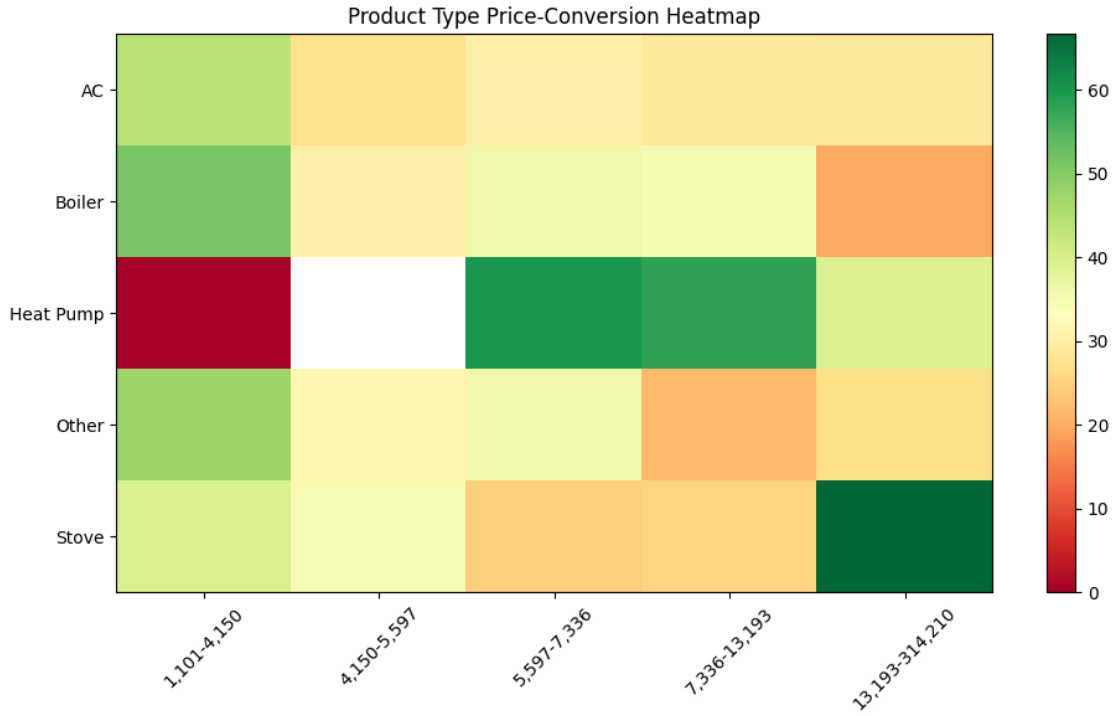
================================================================================
Product Type - Price Conversion Heatmap
================================================================================
Simple pivot table:
price_bin_simple         (1101.459, 4150.254]  (4150.254, 5597.474]  \
main_equipment_category
AC                                  43.750000             27.160494
Boiler                              51.282051             30.215827
Heat Pump                            0.000000                   NaN
Other                               47.647059             31.250000
Stove                               39.393939             34.736842

price_bin_simple         (5597.474, 7336.372]  (7336.372, 13193.412]  \
main_equipment_category
AC                                  30.337079              28.767123
Boiler                              35.714286              35.000000
Heat Pump                           60.000000              58.088235
Other                               35.714286              21.739130
Stove                               24.363636              25.000000
```

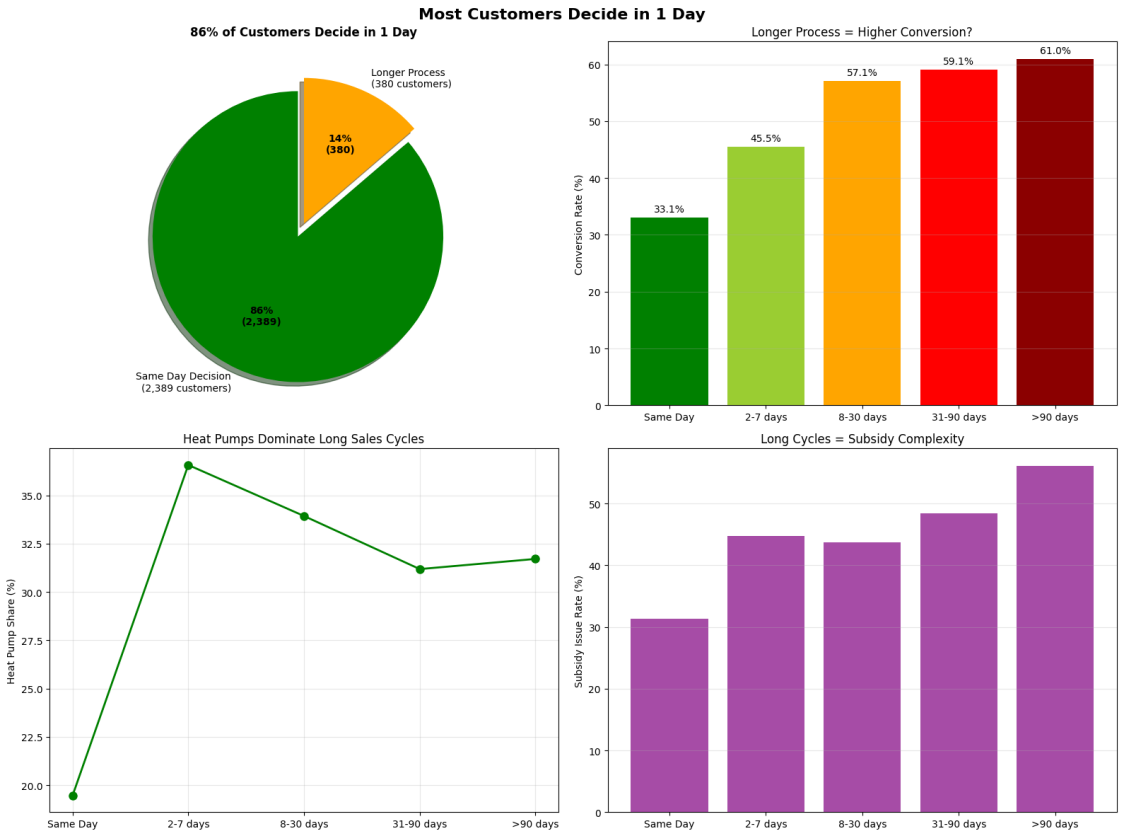
```
price_bin_simple         (13193.412, 314210.61]
main_equipment_category
AC                                     28.571429
Boiler                                 19.607843
Heat Pump                              39.185751
Other                                  26.415094
Stove                                  66.666667
```




```
================================================================================
CHECKING MAIN PRODUCT FAMILY
================================================================================
main_product_family
Chaudière                          926
Poêle                              607
Pompe à chaleur                    523
Climatisation                      495
ECS : Chauffe-eau ou adoucisseur   154
Appareil hybride                    52
Photovoltaïque                      34
Autres                              30
Emetteur de chauffage  ou chappe    25
Produit VMC                         12
Unknown                              9
```

```
Accessoire de pose                    4
Plomberie Sanitaire                   1
Name: count, dtype: int64

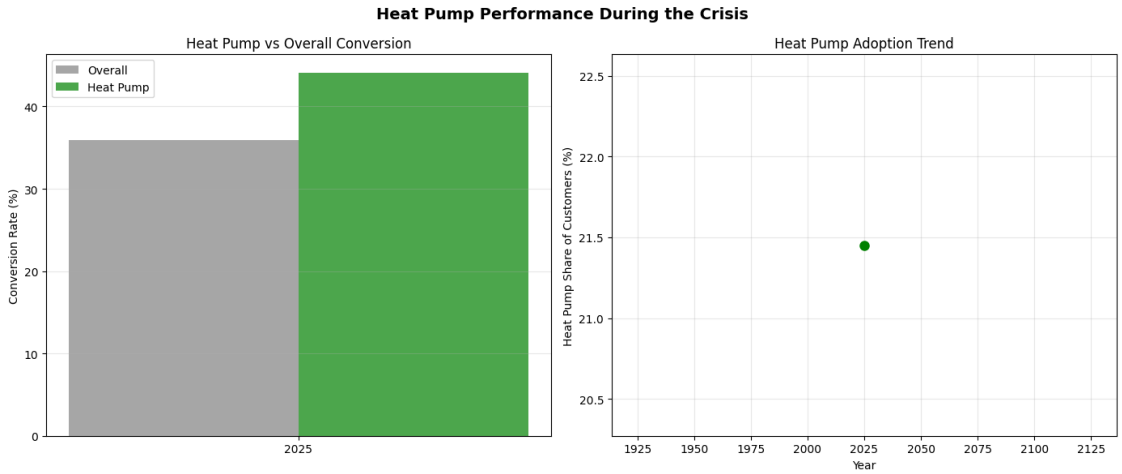
Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263
```




```
================================================================================
Heat Pump Performance Over Time
================================================================================

Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263

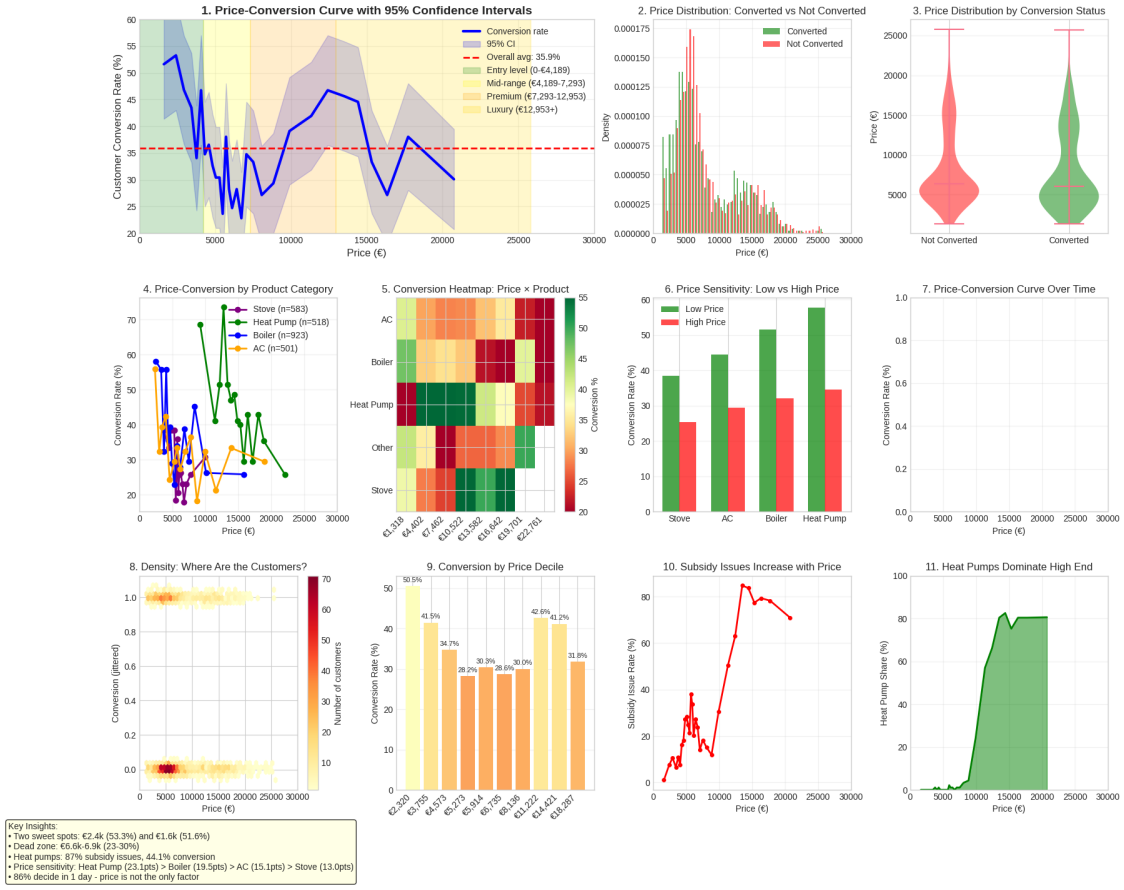
Heat pump customers by year:
        conversion_rate  customer_count  avg_price  subsidy_issue_rate
period
2025              0.441             594  15056.297               0.869
```



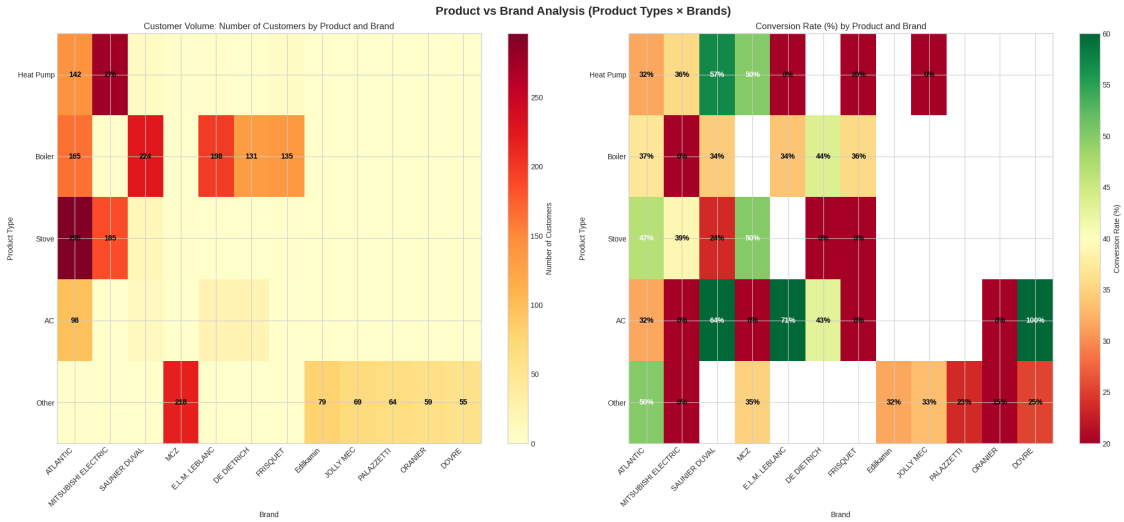

```
Analyzing 2,769 customers
Price range: €1,343 to €25,821
Median price: €6,263

================================================================================
DYNAMIC INSIGHTS CALCULATED:
================================================================================
Sweet spots: €2.4k (53.3%) and €1.6k (51.6%)
Dead zone: €6.6k-6.9k (23-30%)
Heat pumps: 87% subsidy issues, 44.1% conversion
Price sensitivity: Heat Pump (23.1pts) > Boiler (19.5pts) > AC (15.1pts) > Stove
(13.0pts)
Same day decisions: 86%
```

Price-Conversion Dependency Dashboard



Product vs Brand Analysis (Product Types × Brands)



```
================================================================================
PRODUCT × BRAND ANALYSIS
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC  SAUNIER DUVAL  MCZ
E.L.M. LEBLANC  DE DIETRICH  FRISQUET  Edilkamin  JOLLY MEC  PALAZZETTI  ORANIER
DOVRE
main_equipment_category
AC                            142                  276              7    2
2            0         5          0          1           0        0      0
Boiler                        165                    2            224    0
198          131       135          0          0           0        0      0
Heat Pump                     296                  185             17    2
0            1         2          0          0           0        0      0
Other                          98                    1             11    1
21           21         1          0          0           0        1      1
Stove                           2                    1              0  218
0            0         0         79         69          64       59     55

================================================================================
  CONVERSION RATES (%):
--------------------------------------------------------------------------------
main_brand                ATLANTIC  MITSUBISHI ELECTRIC  SAUNIER DUVAL   MCZ
E.L.M. LEBLANC  DE DIETRICH  FRISQUET  Edilkamin  JOLLY MEC  PALAZZETTI  ORANIER
DOVRE
main_equipment_category
AC                           31.7                 35.5           57.1  50.0
0.0          NaN      20.0        NaN        0.0         NaN      NaN    NaN
Boiler                       37.0                  0.0           34.4   NaN
33.8         43.5      35.6        NaN        NaN         NaN      NaN    NaN
Heat Pump                    46.6                 38.9           23.5  50.0
NaN          0.0       0.0        NaN        NaN         NaN      NaN    NaN
Other                        31.6                  0.0           63.6   0.0
71.4         42.9       0.0        NaN        NaN         NaN      0.0  100.0
Stove                        50.0                  0.0            NaN  34.9
NaN          NaN       NaN       31.6       33.3        23.4     15.3   25.5

================================================================================
  BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------

Heat Pump:
  Best brand: MCZ
  Conversion: 50.0%
  Customers: 2
```

```
  Top 3 brands:
    1. MCZ: 50.0% (n=2)
    2. ATLANTIC: 46.6% (n=296)
    3. MITSUBISHI ELECTRIC: 38.9% (n=185)

Boiler:
  Best brand: DE DIETRICH
  Conversion: 43.5%
  Customers: 131
  Top 3 brands:
    1. DE DIETRICH: 43.5% (n=131)
    2. ATLANTIC: 37.0% (n=165)
    3. FRISQUET: 35.6% (n=135)

Stove:
  Best brand: ATLANTIC
  Conversion: 50.0%
  Customers: 2
  Top 3 brands:
    1. ATLANTIC: 50.0% (n=2)
    2. MCZ: 34.9% (n=218)
    3. JOLLY MEC: 33.3% (n=69)

AC:
  Best brand: SAUNIER DUVAL
  Conversion: 57.1%
  Customers: 7
  Top 3 brands:
    1. SAUNIER DUVAL: 57.1% (n=7)
    2. MCZ: 50.0% (n=2)
    3. MITSUBISHI ELECTRIC: 35.5% (n=276)

Other:
  Best brand: DOVRE
  Conversion: 100.0%
  Customers: 1
  Top 3 brands:
    1. DOVRE: 100.0% (n=1)
    2. E.L.M. LEBLANC: 71.4% (n=21)
    3. SAUNIER DUVAL: 63.6% (n=11)

================================================================================
 BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------

ATLANTIC:
  Best product: Stove
  Conversion: 50.0%
```

```
  Customers: 2
  All products:
    Stove: 50.0% (n=2)
    Heat Pump: 46.6% (n=296)
    Boiler: 37.0% (n=165)
    AC: 31.7% (n=142)
    Other: 31.6% (n=98)

MITSUBISHI ELECTRIC:
  Best product: Heat Pump
  Conversion: 38.9%
  Customers: 185
  All products:
    Heat Pump: 38.9% (n=185)
    AC: 35.5% (n=276)
    Boiler: 0.0% (n=2)
    Other: 0.0% (n=1)
    Stove: 0.0% (n=1)

SAUNIER DUVAL:
  Best product: Other
  Conversion: 63.6%
  Customers: 11
  All products:
    Other: 63.6% (n=11)
    AC: 57.1% (n=7)
    Boiler: 34.4% (n=224)
    Heat Pump: 23.5% (n=17)

MCZ:
  Best product: AC
  Conversion: 50.0%
  Customers: 2
  All products:
    AC: 50.0% (n=2)
    Heat Pump: 50.0% (n=2)
    Stove: 34.9% (n=218)
    Other: 0.0% (n=1)

E.L.M. LEBLANC:
  Best product: Other
  Conversion: 71.4%
  Customers: 21
  All products:
    Other: 71.4% (n=21)
    Boiler: 33.8% (n=198)
    AC: 0.0% (n=2)
```

```
DE DIETRICH:
  Best product: Boiler
  Conversion: 43.5%
  Customers: 131
  All products:
    Boiler: 43.5% (n=131)
    Other: 42.9% (n=21)
    Heat Pump: 0.0% (n=1)

FRISQUET:
  Best product: Boiler
  Conversion: 35.6%
  Customers: 135
  All products:
    Boiler: 35.6% (n=135)
    AC: 20.0% (n=5)
    Heat Pump: 0.0% (n=2)
    Other: 0.0% (n=1)

Edilkamin:
  Best product: Stove
  Conversion: 31.6%
  Customers: 79
  All products:
    Stove: 31.6% (n=79)

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Brand Dominance by Product Category:

Heat Pump:
  Volume leader: ATLANTIC (296 customers, 58.8% share)
  Conversion leader: MCZ (50.0%)

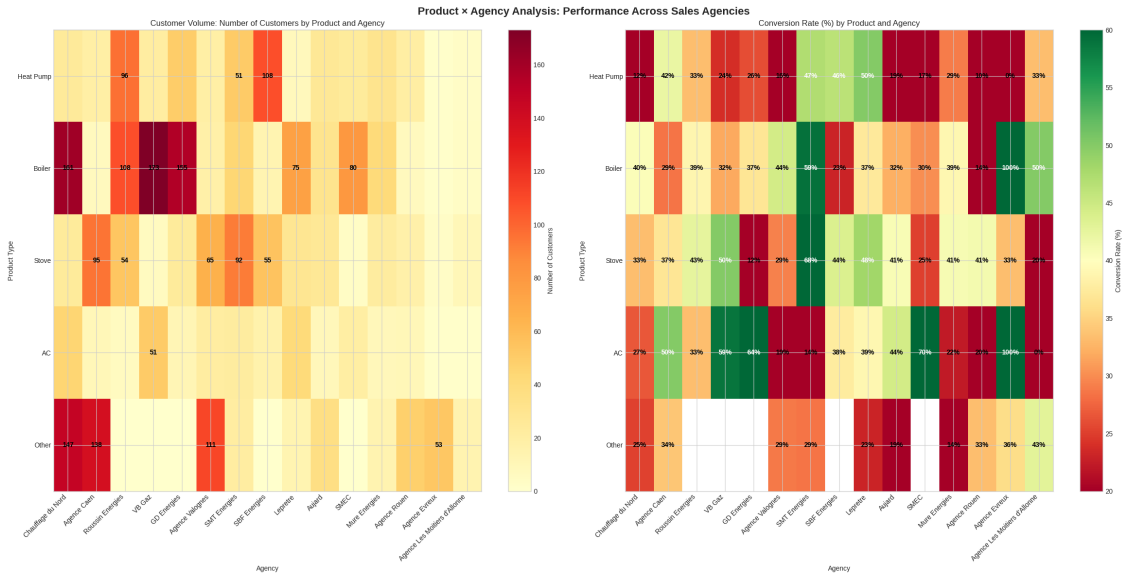
Boiler:
  Volume leader: SAUNIER DUVAL (224 customers, 26.2% share)
  Conversion leader: DE DIETRICH (43.5%)

Stove:
  Volume leader: MCZ (218 customers, 39.9% share)
  Conversion leader: ATLANTIC (50.0%)

AC:
  Volume leader: MITSUBISHI ELECTRIC (276 customers, 63.4% share)
  Conversion leader: SAUNIER DUVAL (57.1%)
```

```
Other:
  Volume leader: ATLANTIC (98 customers, 62.8% share)
  Conversion leader: DOVRE (100.0%)

Top 15 agencies by customer volume:
  Chauffage du Nord: 402 customers, 31.1% conversion
  Agence Caen: 269 customers, 36.1% conversion
  Roussin Energies: 264 customers, 37.5% conversion
  VB Gaz: 251 customers, 37.1% conversion
  GD Energies: 240 customers, 33.3% conversion
  Agence Valognes: 234 customers, 28.2% conversion
  SMT Energies: 230 customers, 53.0% conversion
  SBF Energies: 192 customers, 43.2% conversion
  Lepretre: 166 customers, 39.2% conversion
  Aujard: 130 customers, 28.5% conversion
  SMEC: 128 customers, 33.6% conversion
  Mure Energies: 117 customers, 32.5% conversion
  Agence Rouen: 102 customers, 27.5% conversion
  Agence Evreux: 59 customers, 37.3% conversion
  Agence Les Moitiers d'Allonne: 32 customers, 34.4% conversion
```




```
================================================================================
PRODUCT × AGENCY ANALYSIS
================================================================================

  CUSTOMER VOLUME BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
```

```
main_agency              Chauffage du Nord  Agence Caen  Roussin Energies  VB
Gaz  GD Energies  Agence Valognes  SMT Energies  SBF Energies  Lepretre  Aujard
SMEC  Mure Energies  Agence Rouen  Agence Evreux  Agence Les Moitiers d'Allonne
main_equipment_category
AC                                      25           19                 96
21           50               19            51           108         8      27
24             31            20              1                              3
Boiler                                 161            7                108
173          155               18            44            13        75      31
80             41             7              1                              4
Heat Pump                               24           95                 54
6           24               65            92            55        29      27
4             22            17              3                             10
Other                                   45           10                  6
51           11               21            22            16        41       9
20              9            10              1                              1
Stove                                  147          138                  0
0            0              111            21             0        13      36
0             14            48             53                             14

================================================================================
 CONVERSION RATES (%) BY PRODUCT AND AGENCY:
--------------------------------------------------------------------------------
--------------------
main_agency              Chauffage du Nord  Agence Caen  Roussin Energies  VB
Gaz  GD Energies  Agence Valognes  SMT Energies  SBF Energies  Lepretre  Aujard
SMEC  Mure Energies  Agence Rouen  Agence Evreux  Agence Les Moitiers d'Allonne
main_equipment_category
AC                                    12.0         42.1              33.3
23.8         26.0             15.8          47.1          46.3      50.0    18.5
16.7           29.0          10.0            0.0                            33.3
Boiler                                40.4         28.6              38.9
31.8         36.8             44.4          59.1          23.1      37.3    32.3
30.0           39.0          14.3          100.0                            50.0
Heat Pump                             33.3         36.8              42.6
50.0         12.5             29.2          68.5          43.6      48.3    40.7
25.0           40.9          41.2           33.3                            20.0
Other                                 26.7         50.0              33.3
58.8         63.6             19.0          13.6          37.5      39.0    44.4
70.0           22.2          20.0          100.0                             0.0
Stove                                 25.2         34.1               NaN
NaN          NaN              28.8          28.6           NaN      23.1    19.4
NaN            14.3          33.3           35.8                            42.9

================================================================================
 BEST AGENCY BY PRODUCT TYPE
--------------------------------------------------------------------------------
--------------------
```

```
Heat Pump:
  Best agency: SMT Energies
  Conversion: 68.5%
  Customers: 92
  Top 3 agencies:
    1. SMT Energies: 68.5% (n=92)
    2. VB Gaz: 50.0% (n=6)
    3. Lepretre: 48.3% (n=29)

Boiler:
  Best agency: Agence Evreux
  Conversion: 100.0%
  Customers: 1
  Top 3 agencies:
    1. Agence Evreux: 100.0% (n=1)
    2. SMT Energies: 59.1% (n=44)
    3. Agence Les Moitiers d'Allonne: 50.0% (n=4)

Stove:
  Best agency: Agence Les Moitiers d'Allonne
  Conversion: 42.9%
  Customers: 14
  Top 3 agencies:
    1. Agence Les Moitiers d'Allonne: 42.9% (n=14)
    2. Agence Evreux: 35.8% (n=53)
    3. Agence Caen: 34.1% (n=138)

AC:
  Best agency: Lepretre
  Conversion: 50.0%
  Customers: 8
  Top 3 agencies:
    1. Lepretre: 50.0% (n=8)
    2. SMT Energies: 47.1% (n=51)
    3. SBF Energies: 46.3% (n=108)

Other:
  Best agency: Agence Evreux
  Conversion: 100.0%
  Customers: 1
  Top 3 agencies:
    1. Agence Evreux: 100.0% (n=1)
    2. SMEC: 70.0% (n=20)
    3. GD Energies: 63.6% (n=11)

================================================================================
 BEST PRODUCT BY AGENCY
```

```
--------------------------------------------------------------------------------
--------------------

Chauffage du Nord:
  Best product: Boiler
  Conversion: 40.4%
  Customers: 161
  All products:
    Boiler: 40.4% (n=161)
    Heat Pump: 33.3% (n=24)
    Other: 26.7% (n=45)
    Stove: 25.2% (n=147)
    AC: 12.0% (n=25)

Agence Caen:
  Best product: Other
  Conversion: 50.0%
  Customers: 10
  All products:
    Other: 50.0% (n=10)
    AC: 42.1% (n=19)
    Heat Pump: 36.8% (n=95)
    Stove: 34.1% (n=138)
    Boiler: 28.6% (n=7)

Roussin Energies:
  Best product: Heat Pump
  Conversion: 42.6%
  Customers: 54
  All products:
    Heat Pump: 42.6% (n=54)
    Boiler: 38.9% (n=108)
    AC: 33.3% (n=96)
    Other: 33.3% (n=6)

VB Gaz:
  Best product: Other
  Conversion: 58.8%
  Customers: 51
  All products:
    Other: 58.8% (n=51)
    Heat Pump: 50.0% (n=6)
    Boiler: 31.8% (n=173)
    AC: 23.8% (n=21)

GD Energies:
  Best product: Other
  Conversion: 63.6%
```

```
  Customers: 11
  All products:
    Other: 63.6% (n=11)
    Boiler: 36.8% (n=155)
    AC: 26.0% (n=50)
    Heat Pump: 12.5% (n=24)

Agence Valognes:
  Best product: Boiler
  Conversion: 44.4%
  Customers: 18
  All products:
    Boiler: 44.4% (n=18)
    Heat Pump: 29.2% (n=65)
    Stove: 28.8% (n=111)
    Other: 19.0% (n=21)
    AC: 15.8% (n=19)

SMT Energies:
  Best product: Heat Pump
  Conversion: 68.5%
  Customers: 92
  All products:
    Heat Pump: 68.5% (n=92)
    Boiler: 59.1% (n=44)
    AC: 47.1% (n=51)
    Stove: 28.6% (n=21)
    Other: 13.6% (n=22)

SBF Energies:
  Best product: AC
  Conversion: 46.3%
  Customers: 108
  All products:
    AC: 46.3% (n=108)
    Heat Pump: 43.6% (n=55)
    Other: 37.5% (n=16)
    Boiler: 23.1% (n=13)

Lepretre:
  Best product: AC
  Conversion: 50.0%
  Customers: 8
  All products:
    AC: 50.0% (n=8)
    Heat Pump: 48.3% (n=29)
    Other: 39.0% (n=41)
    Boiler: 37.3% (n=75)
```

```
    Stove: 23.1% (n=13)

Aujard:
  Best product: Other
  Conversion: 44.4%
  Customers: 9
  All products:
    Other: 44.4% (n=9)
    Heat Pump: 40.7% (n=27)
    Boiler: 32.3% (n=31)
    Stove: 19.4% (n=36)
    AC: 18.5% (n=27)

================================================================================
 AGENCY PERFORMANCE SUMMARY
================================================================================

Top 15 agencies by volume:
                              conversion_rate  customer_count  heat_pump_pct
subsidy_issue_pct  avg_price
main_agency
Chauffage du Nord                        31.1             402            7.2
44.3   6283.317
Agence Caen                              36.1             269           36.4
50.2   9417.039
Roussin Energies                         37.5             264           22.7
22.3   8952.811
VB Gaz                                   37.1             251            4.4
3.6   5430.851
GD Energies                              33.3             240           15.4
13.8   6256.769
Agence Valognes                          28.2             234           32.1
47.4   8831.968
SMT Energies                             53.0             230           45.7
57.8   8078.604
SBF Energies                             43.2             192           32.8
28.6  11685.687
Lepretre                                 39.2             166           18.7
33.7   7931.980
Aujard                                   28.5             130           23.1
20.8   8156.938
SMEC                                     33.6             128            5.5
10.2   5963.468
Mure Energies                            32.5             117           23.1
25.6   7866.604
Agence Rouen                             27.5             102           18.6
47.1   8879.362
Agence Evreux                            37.3              59            5.1
```

```
28.8   6706.337
Agence Les Moitiers d'Allonne             34.4              32           31.2
40.6   9250.438

================================================================================
  STRATEGIC INSIGHTS
================================================================================

Agency Excellence by Product Category:

Heat Pump:
  Conversion leader: SMT Energies (68.5%, n=92)

Boiler:
  Conversion leader: Agence Evreux (100.0%, n=1)

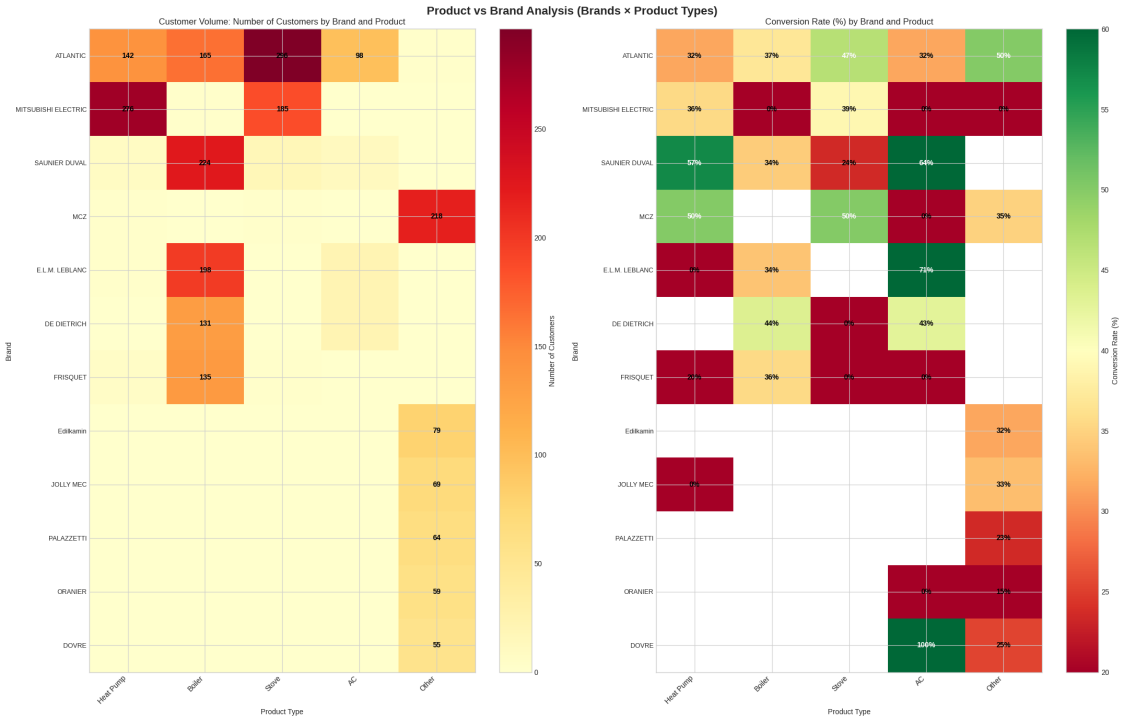
Stove:
  Conversion leader: Agence Les Moitiers d'Allonne (42.9%, n=14)

AC:
  Conversion leader: Lepretre (50.0%, n=8)

Other:
  Conversion leader: Agence Evreux (100.0%, n=1)

================================================================================
  HEAT PUMP SPECIALISTS
================================================================================

Top 5 agencies for Heat Pump conversion:
  SMT Energies: 68.5% (n=92)
  VB Gaz: 50.0% (n=6)
  Lepretre: 48.3% (n=29)
  SBF Energies: 43.6% (n=55)
  Roussin Energies: 42.6% (n=54)
```




```
================================================================================
BRAND × PRODUCT ANALYSIS (Swapped Axes)
================================================================================

  CUSTOMER VOLUME (Number of Customers):
--------------------------------------------------------------------------------
main_equipment_category   AC  Boiler  Heat Pump  Other  Stove
main_brand
ATLANTIC                 142     165        296     98      2
MITSUBISHI ELECTRIC      276       2        185      1      1
SAUNIER DUVAL              7     224         17     11      0
MCZ                        2       0          2      1    218
E.L.M. LEBLANC             2     198          0     21      0
DE DIETRICH                0     131          1     21      0
FRISQUET                   5     135          2      1      0
Edilkamin                  0       0          0      0     79
JOLLY MEC                  1       0          0      0     69
PALAZZETTI                 0       0          0      0     64
ORANIER                    0       0          0      1     59
DOVRE                      0       0          0      1     55

================================================================================
  CONVERSION RATES (%):
```

```
--------------------------------------------------------------------------------
main_equipment_category    AC  Boiler  Heat Pump  Other  Stove
main_brand
ATLANTIC                 31.7    37.0       46.6   31.6   50.0
MITSUBISHI ELECTRIC      35.5     0.0       38.9    0.0    0.0
SAUNIER DUVAL            57.1    34.4       23.5   63.6    NaN
MCZ                      50.0     NaN       50.0    0.0   34.9
E.L.M. LEBLANC            0.0    33.8        NaN   71.4    NaN
DE DIETRICH               NaN    43.5        0.0   42.9    NaN
FRISQUET                 20.0    35.6        0.0    0.0    NaN
Edilkamin                 NaN     NaN        NaN    NaN   31.6
JOLLY MEC                 0.0     NaN        NaN    NaN   33.3
PALAZZETTI                NaN     NaN        NaN    NaN   23.4
ORANIER                   NaN     NaN        NaN    0.0   15.3
DOVRE                     NaN     NaN        NaN  100.0   25.5

================================================================================
  BEST PRODUCT BY BRAND
--------------------------------------------------------------------------------

ATLANTIC:
  Best product: Stove
  Conversion: 50.0%
  Customers: 2
  Top 3 products:
    1. Stove: 50.0% (n=2)
    2. Heat Pump: 46.6% (n=296)
    3. Boiler: 37.0% (n=165)

MITSUBISHI ELECTRIC:
  Best product: Heat Pump
  Conversion: 38.9%
  Customers: 185
  Top 3 products:
    1. Heat Pump: 38.9% (n=185)
    2. AC: 35.5% (n=276)
    3. Boiler: 0.0% (n=2)

SAUNIER DUVAL:
  Best product: Other
  Conversion: 63.6%
  Customers: 11
  Top 3 products:
    1. Other: 63.6% (n=11)
    2. AC: 57.1% (n=7)
    3. Boiler: 34.4% (n=224)

MCZ:
```

```
  Best product: AC
  Conversion: 50.0%
  Customers: 2
  Top 3 products:
    1. AC: 50.0% (n=2)
    2. Heat Pump: 50.0% (n=2)
    3. Stove: 34.9% (n=218)

E.L.M. LEBLANC:
  Best product: Other
  Conversion: 71.4%
  Customers: 21
  Top 3 products:
    1. Other: 71.4% (n=21)
    2. Boiler: 33.8% (n=198)
    3. AC: 0.0% (n=2)

DE DIETRICH:
  Best product: Boiler
  Conversion: 43.5%
  Customers: 131
  Top 3 products:
    1. Boiler: 43.5% (n=131)
    2. Other: 42.9% (n=21)
    3. Heat Pump: 0.0% (n=1)

FRISQUET:
  Best product: Boiler
  Conversion: 35.6%
  Customers: 135
  Top 3 products:
    1. Boiler: 35.6% (n=135)
    2. AC: 20.0% (n=5)
    3. Heat Pump: 0.0% (n=2)

Edilkamin:
  Best product: Stove
  Conversion: 31.6%
  Customers: 79
  Top 3 products:
    1. Stove: 31.6% (n=79)

JOLLY MEC:
  Best product: Stove
  Conversion: 33.3%
  Customers: 69
  Top 3 products:
    1. Stove: 33.3% (n=69)
```

```
    2. AC: 0.0% (n=1)

PALAZZETTI:
  Best product: Stove
  Conversion: 23.4%
  Customers: 64
  Top 3 products:
    1. Stove: 23.4% (n=64)

================================================================================
 BEST BRAND BY PRODUCT TYPE
--------------------------------------------------------------------------------

Heat Pump:
  Best brand: MCZ
  Conversion: 50.0%
  Customers: 2
  Top 5 brands:
    1. MCZ: 50.0% (n=2)
    2. ATLANTIC: 46.6% (n=296)
    3. MITSUBISHI ELECTRIC: 38.9% (n=185)
    4. SAUNIER DUVAL: 23.5% (n=17)
    5. DE DIETRICH: 0.0% (n=1)

Boiler:
  Best brand: DE DIETRICH
  Conversion: 43.5%
  Customers: 131
  Top 5 brands:
    1. DE DIETRICH: 43.5% (n=131)
    2. ATLANTIC: 37.0% (n=165)
    3. FRISQUET: 35.6% (n=135)
    4. SAUNIER DUVAL: 34.4% (n=224)
    5. E.L.M. LEBLANC: 33.8% (n=198)

Stove:
  Best brand: ATLANTIC
  Conversion: 50.0%
  Customers: 2
  Top 5 brands:
    1. ATLANTIC: 50.0% (n=2)
    2. MCZ: 34.9% (n=218)
    3. JOLLY MEC: 33.3% (n=69)
    4. Edilkamin: 31.6% (n=79)
    5. DOVRE: 25.5% (n=55)

AC:
  Best brand: SAUNIER DUVAL
```

```
  Conversion: 57.1%
  Customers: 7
  Top 5 brands:
    1. SAUNIER DUVAL: 57.1% (n=7)
    2. MCZ: 50.0% (n=2)
    3. MITSUBISHI ELECTRIC: 35.5% (n=276)
    4. ATLANTIC: 31.7% (n=142)
    5. FRISQUET: 20.0% (n=5)

Other:
  Best brand: DOVRE
  Conversion: 100.0%
  Customers: 1
  Top 5 brands:
    1. DOVRE: 100.0% (n=1)
    2. E.L.M. LEBLANC: 71.4% (n=21)
    3. SAUNIER DUVAL: 63.6% (n=11)
    4. DE DIETRICH: 42.9% (n=21)
    5. ATLANTIC: 31.6% (n=98)

================================================================================
 STRATEGIC INSIGHTS
================================================================================

Brand Performance by Product Category:

ATLANTIC:
  Stove: 50.0% (n=2)
  Heat Pump: 46.6% (n=296)
  Boiler: 37.0% (n=165)

MITSUBISHI ELECTRIC:
  Heat Pump: 38.9% (n=185)
  AC: 35.5% (n=276)
  Boiler: 0.0% (n=2)

SAUNIER DUVAL:
  Other: 63.6% (n=11)
  AC: 57.1% (n=7)
  Boiler: 34.4% (n=224)

MCZ:
  AC: 50.0% (n=2)
  Heat Pump: 50.0% (n=2)
  Stove: 34.9% (n=218)

E.L.M. LEBLANC:
  Other: 71.4% (n=21)
```

```
  Boiler: 33.8% (n=198)
  AC: 0.0% (n=2)

DE DIETRICH:
  Boiler: 43.5% (n=131)
  Other: 42.9% (n=21)
  Heat Pump: 0.0% (n=1)

FRISQUET:
  Boiler: 35.6% (n=135)
  AC: 20.0% (n=5)
  Heat Pump: 0.0% (n=2)

Edilkamin:
  Stove: 31.6% (n=79)

================================================================================
 MARKET LEADERS BY PRODUCT CATEGORY
================================================================================

Heat Pump:
  Volume leader: ATLANTIC (296 customers, 58.8% share)
  Conversion leader: MCZ (50.0%)

Boiler:
  Volume leader: SAUNIER DUVAL (224 customers, 26.2% share)
  Conversion leader: DE DIETRICH (43.5%)

Stove:
  Volume leader: MCZ (218 customers, 39.9% share)
  Conversion leader: ATLANTIC (50.0%)

AC:
  Volume leader: MITSUBISHI ELECTRIC (276 customers, 63.4% share)
  Conversion leader: SAUNIER DUVAL (57.1%)

Other:
  Volume leader: ATLANTIC (98 customers, 62.8% share)
  Conversion leader: DOVRE (100.0%)
```

(main_equipment_category	AC	Boiler	Heat Pump	Other	Stove
main_brand					
ATLANTIC	142	165	296	98	2
MITSUBISHI ELECTRIC	276	2	185	1	1
SAUNIER DUVAL	7	224	17	11	0
MCZ	2	0	2	1	218
E.L.M. LEBLANC	2	198	0	21	0

```
DE DIETRICH                0     131          1     21      0
FRISQUET                   5     135          2      1      0
Edilkamin                  0       0          0      0     79
JOLLY MEC                  1       0          0      0     69
PALAZZETTI                 0       0          0      0     64
ORANIER                    0       0          0      1     59
DOVRE                      0       0          0      1     55,
main_equipment_category         AC     Boiler  Heat Pump       Other  \
main_brand
ATLANTIC                 31.690141  36.969697  46.621622   31.632653
MITSUBISHI ELECTRIC      35.507246   0.000000  38.918919    0.000000
SAUNIER DUVAL            57.142857  34.375000  23.529412   63.636364
MCZ                      50.000000        NaN  50.000000    0.000000
E.L.M. LEBLANC            0.000000  33.838384        NaN   71.428571
DE DIETRICH                    NaN  43.511450   0.000000   42.857143
FRISQUET                 20.000000  35.555556   0.000000    0.000000
Edilkamin                      NaN        NaN        NaN         NaN
JOLLY MEC                 0.000000        NaN        NaN         NaN
PALAZZETTI                     NaN        NaN        NaN         NaN
ORANIER                        NaN        NaN        NaN    0.000000
DOVRE                          NaN        NaN        NaN  100.000000

main_equipment_category      Stove
main_brand
ATLANTIC                 50.000000
MITSUBISHI ELECTRIC       0.000000
SAUNIER DUVAL                  NaN
MCZ                      34.862385
E.L.M. LEBLANC                 NaN
DE DIETRICH                    NaN
FRISQUET                       NaN
Edilkamin                31.645570
JOLLY MEC                33.333333
PALAZZETTI               23.437500
ORANIER                  15.254237
DOVRE                    25.454545  )
```

```python
import ipynbname
import subprocess
import os
import sys

def export_notebook_to_pdf(overwrite=True, verbose=True):
    """
    Export current notebook to PDF with overwrite control

    Args:
```

```
        overwrite (bool): If True, overwrite existing PDF file
        verbose (bool): If True, print progress messages

    Returns:
        bool: True if successful, False otherwise
    """
    try:
        # Get notebook path
        notebook_path = ipynbname.path()
        notebook_path_str = str(notebook_path)
        pdf_path = notebook_path_str.replace('.ipynb', '.pdf')

        if verbose:
            print(f"  Notebook: {notebook_path.name}")
            print(f"  PDF path: {pdf_path}")

        # Check if PDF exists
        if os.path.exists(pdf_path):
            if overwrite:
                if verbose:
                    print(f"  Removing existing PDF: {pdf_path}")
                os.remove(pdf_path)
            else:
                if verbose:
                    print(f"  PDF already exists and overwrite=False")
                return False

        # Convert to PDF
        if verbose:
            print("  Converting to PDF...")

        result = subprocess.run([
            'jupyter', 'nbconvert', '--to', 'pdf',
            '--no-prompt',  # Suppress input prompts
            notebook_path_str
        ], capture_output=True, text=True)

        if result.returncode == 0:
            if verbose:
                print(f"  PDF saved: {pdf_path}")
            return True
        else:
            if verbose:
                print(f"  Conversion failed:")
                print(result.stderr)
            return False
```

```python
    except Exception as e:
        if verbose:
            print(f"  Error: {e}")
        return False

# Use the function
if __name__ == "__main__":
    # Overwrite existing PDF
    export_notebook_to_pdf(overwrite=True)

    # Or create new version without overwriting
    # export_notebook_to_pdf(overwrite=False)export_full_notebook_to_pdf()
```

```
 Notebook: _eda_price_2025_Q4.ipynb
 PDF path: /home/valeriya/project/homeserve/ai-france-
hvac/_eda_price_2025_Q4.pdf
 Removing existing PDF: /home/valeriya/project/homeserve/ai-france-
hvac/_eda_price_2025_Q4.pdf
 Converting to PDF…
 PDF saved: /home/valeriya/project/homeserve/ai-france-
hvac/_eda_price_2025_Q4.pdf
```